Chapter 1

Snow products

1.1 Study area

This thesis focuses on the Italian hydrological portion of the Greater Alpine Region (GAR), comprising all Italian territory located north of 43 °N together with the Swiss portion of the Po basin. The main features of this territory are respectively:

- *the Italian Alps*, the southernmost part of the Alpine Arc. They extend for approximately 1200 km, covering an area of about 50 000 km^2. The highest peak, Monte Bianco, reaches 4810 m a.s.l., and summits above 3000 m a.s.l. are fairly common throughout the range. The considerable elevations sustain 901 glaciers, whose total surface area is roughly 368 km^2 (Smiraglia and Diolaiuti 2016).

- *the Po Plain*, a predominantly flat region shaped by an initial phase of marine sedimentation followed by later fluvial deposition. It is drained by the Po river, Italy's longest, which flows for approximately 652 km. Together with the Veneto-Friulian plains, it forms one of the country's principal agricultural and industrial areas, home to roughly 20 million people.

- *the Northern Apennines*, formed by the collision between the western margin of the African Promontory and the Sardinia-Corsica block. Their highest peak reaches only 2165 m a.s.l., insufficient to prevent the complete melt of the seasonal snowpack (Soldati and Marchetti 2018).

The complex orography of the study area, examined in greater detail in Figure 1.1, governs its climate. Indeed, the Italian Alps act as a barrier against cold air masses from the north, while the Apennines shield the Po Valley from the milder influence of the Tyrrhenian Sea. The main climate types found within in the study area are:

- *the Alpine climate*, dominant across the Alps and the Northern Apennines, associated with low temperatures both at night and during winter, and with a summer peak in precipitation.

- *the Po Valley climate*, characterized by cold winters and hot summers, with precipitation peaking in spring and autumn (Ministry for the Environment, Land and Sea 2007).

The high elevations of the Alpine Range, combined with the low winter temperatures, allow the Italian Alps to store substantial volumes of water in solid form, often exceeding 10 Gm^3 (Avanzi et al. 2023). Some snowfall also occurs over the Norther Apennines, though in far smaller amounts. These differences in snow cover give rise to distinct hydrological regimes, as up to 50% of the annual stream flow of the main rivers originating in the Italian Alps consists of meltwater (Colombo et al. 2023), and peak discharge is reached in early summer when ablation is most intense. On the other hand, rivers draining the Northern Apennines rely predominantly on rainfall, and often experience low flow, or even run completely dry, during summer.

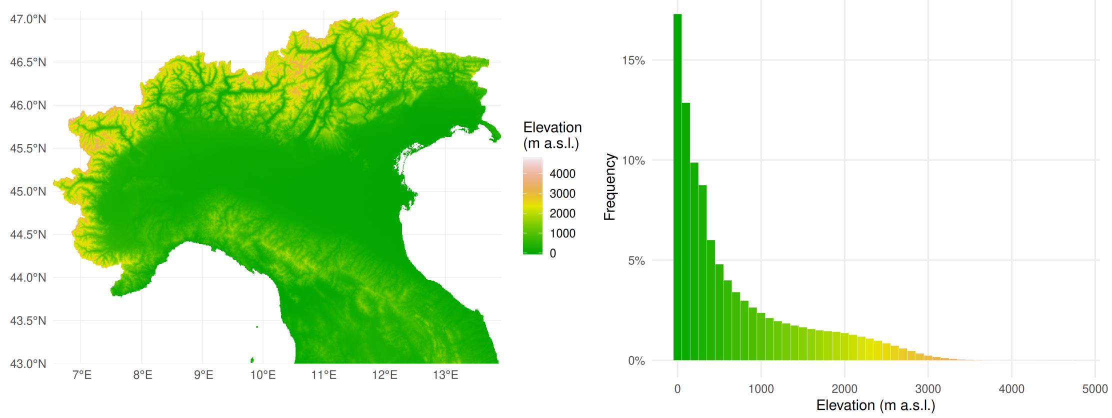

Figure 1.1: Topography and elevation distribution of the study area. The frequency histogram is based on 100 meters wide elevation bins. Elevation data were obtained from ALOS AW3D30 global digital elevation model (DEM).

1.2 IT-Snow

IT-SNOW is a high resolution, multi-year snow reanalysis dataset covering the whole Italian territory. It provides daily maps of snow water equivalent (SWE), snow depth, bulk snow density and liquid water content of the snowpack for a period spanning from September 1^{st}, 2010 to August 31^{st}, 2025. The spatial resolution is approximately 500 meters. IT-SNOW maps are the output of an operational chain developed by CIMA Research Foundation called S3M Italy.

1.2.1 S3M

S3M, also known as *Snow Multidata Mapping and Modelling*, is a hydrology-oriented cryospheric model that simulates seasonal snow evolution through time. S3M is a raster-based model treating pixels as independent units, as it neglects mass and energy transport processes such as wind redistribution. The same set of ordinary differential equations (ODEs) is solved for each grid point by means of the Euler method: the hourly time-step can be adjusted by the user to meet specific application needs. All input and output variables are distributed.

S3M is specifically designed for large-scale, real-time application in order to be an useful tool in flood forecasting frameworks. This model achieves a high level of performance by focusing on key snow evolution drivers via precipitation-phase partitioning and a hybrid temperature and radiation-driven melt parametrization.

1.2.2 Results

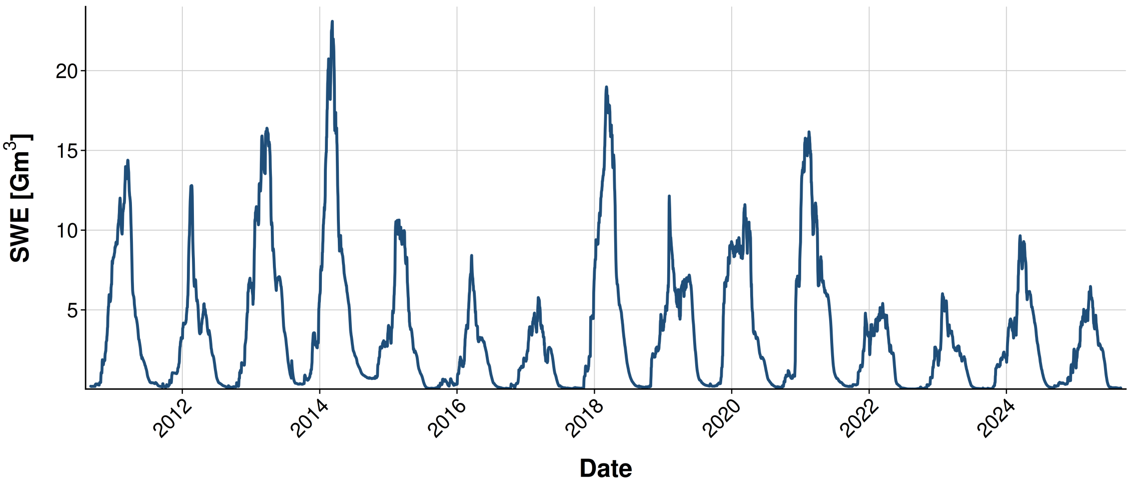

Figure 1.2: Total SWE evolution

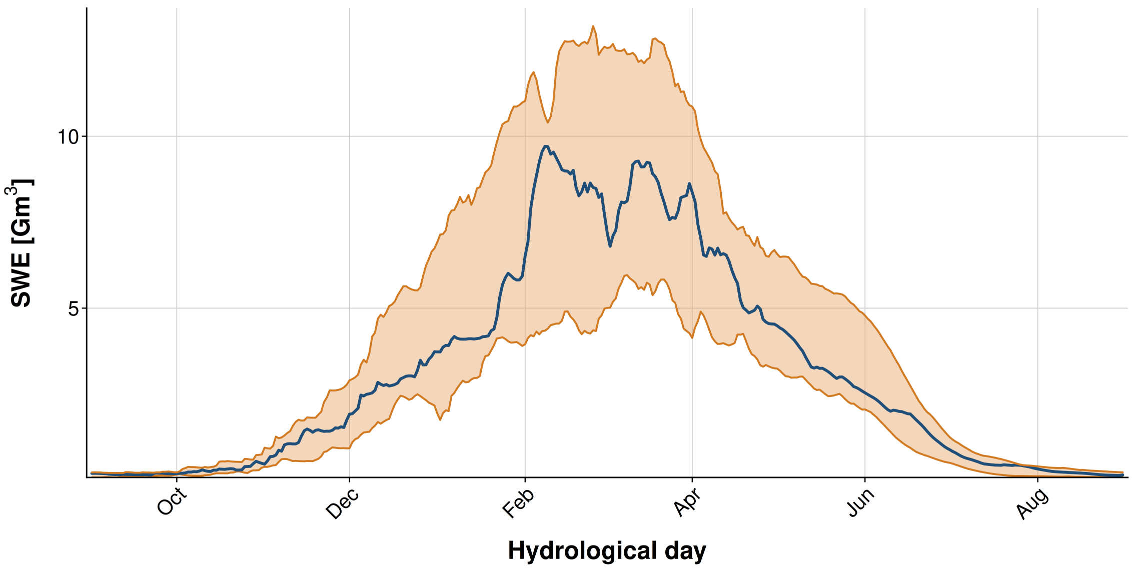

Figure 1.3: Total SWE climatologies.

1.3 MODIS

NASA's Moderate Resolution Imaging Spectroradiometer (MODIS) is a major tool in mapping snow cover extent and duration thanks to the detailed spectral coverage of Earth surface it provides. This tool has been used in order to assess snow cover duration (SCD) trends across various regions the Greater Alpine Region (Fugazza et al. 2021) or the Karakoram (Almagioni et al. 2025). One of the latest studies on snow cover metric characteristics and trends focuses on the Italian territory (Manara et al. 2025), and therefore on the Italian Alps, which are the primary focus of this thesis.

1.3.1 Dataset Generation and Gap-Filling Strategy

In its latest version, this Italian snow cover dataset spans from 2000 to 2025 and consists of binary maps, where a value of zero represents to bare ground, whereas one is associated only to snow covered pixels. Available on a daily basis, these maps feature a spatial resolution of roughly 500 meters.

The starting point of the workline that led the authors to those maps are the raw modis datas. To correctly identify snow covered pixels advantage is taken on a property of snow, which is that it has very high reflectance in the visible part of the electromagnetic spectrum, whereas it absorbs most of the shortwave radiation. This characteristics is highlighted by the normalized difference snow index (NDSI), which is defined as where band 4 is in the visible range (in particular $0.545 - 0.565\,\mu$m) and band 6 (or 7) is in the shortwave infrared (namely $1.628 - 1.672\,\mu$m for Terra and $2.105 - 2.155\,\mu$m for Aqua).

1.3.2 Results

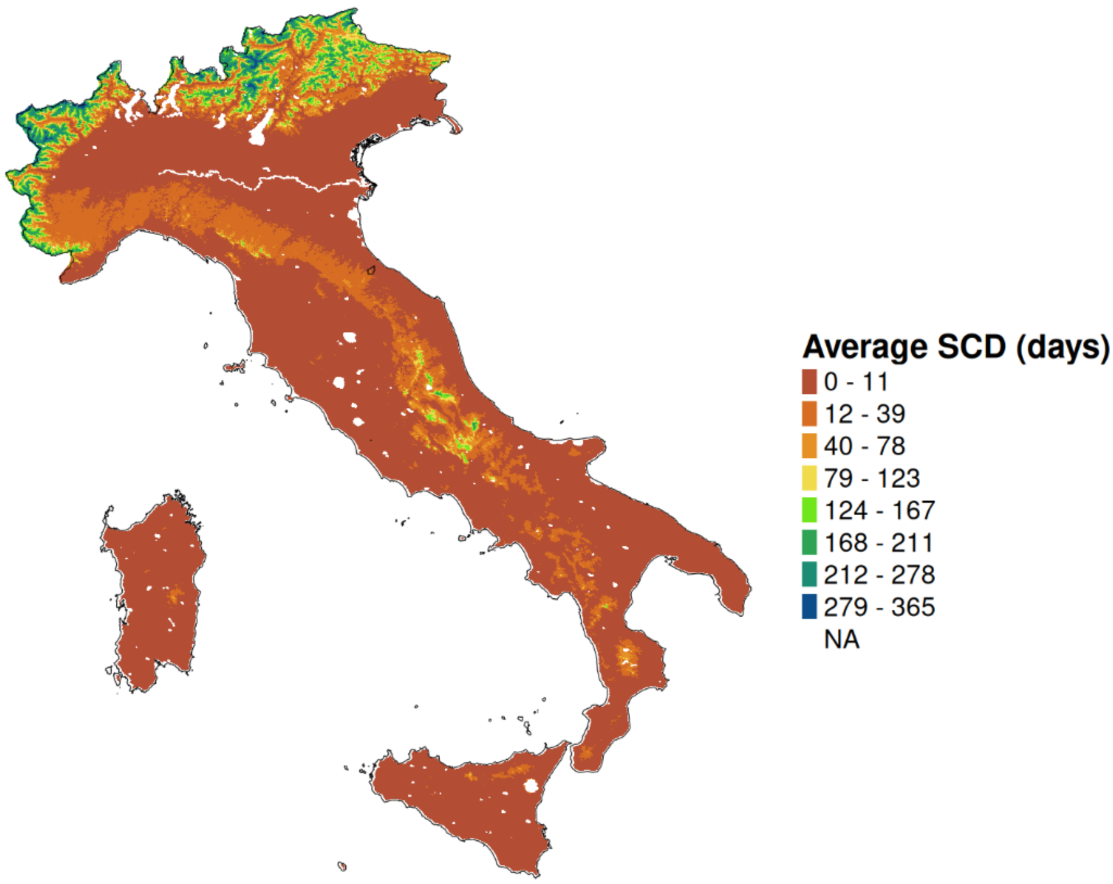

Figure 1.4: MODIS

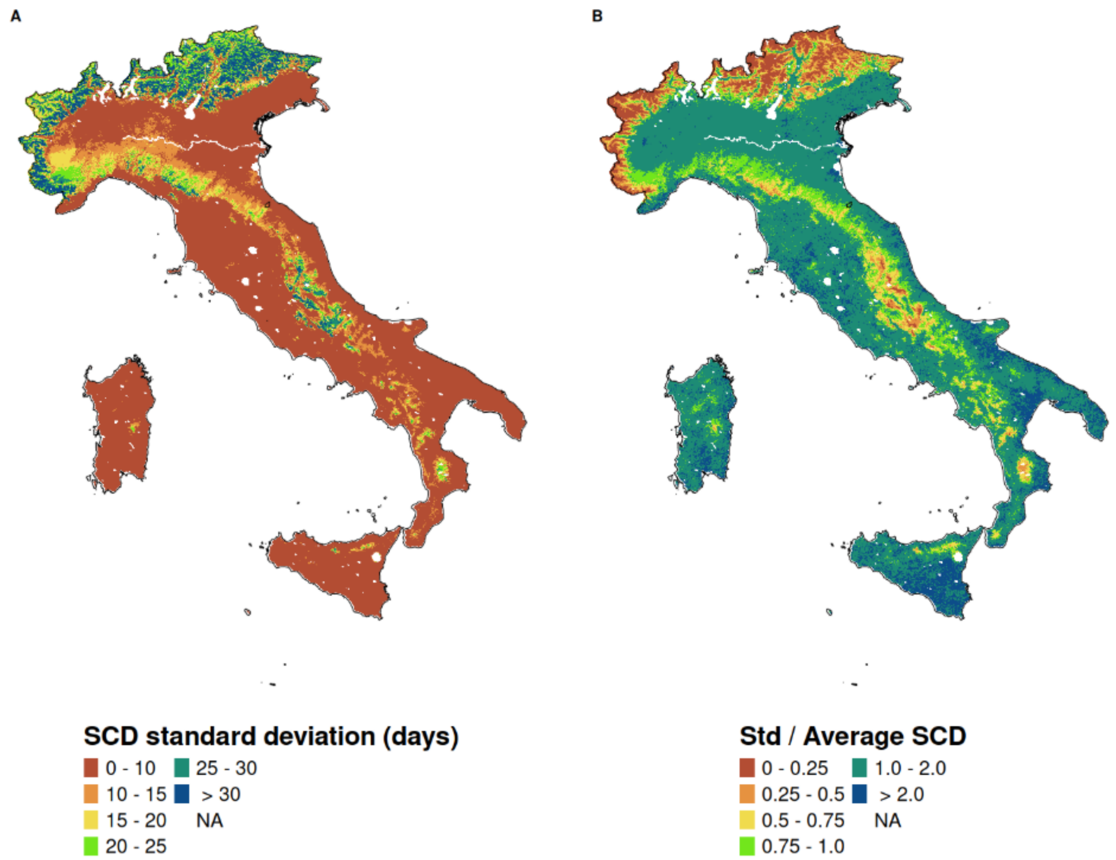

Figure 1.5: MODIS

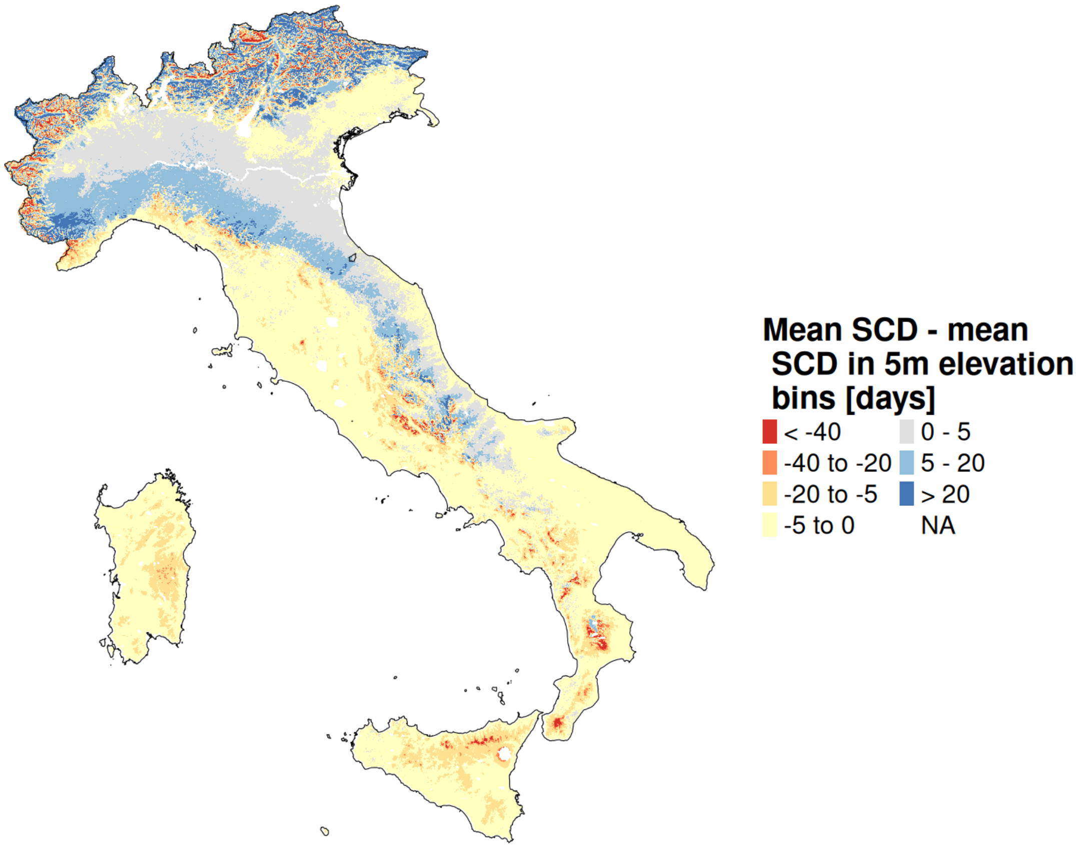

Figure 1.6: MODIS

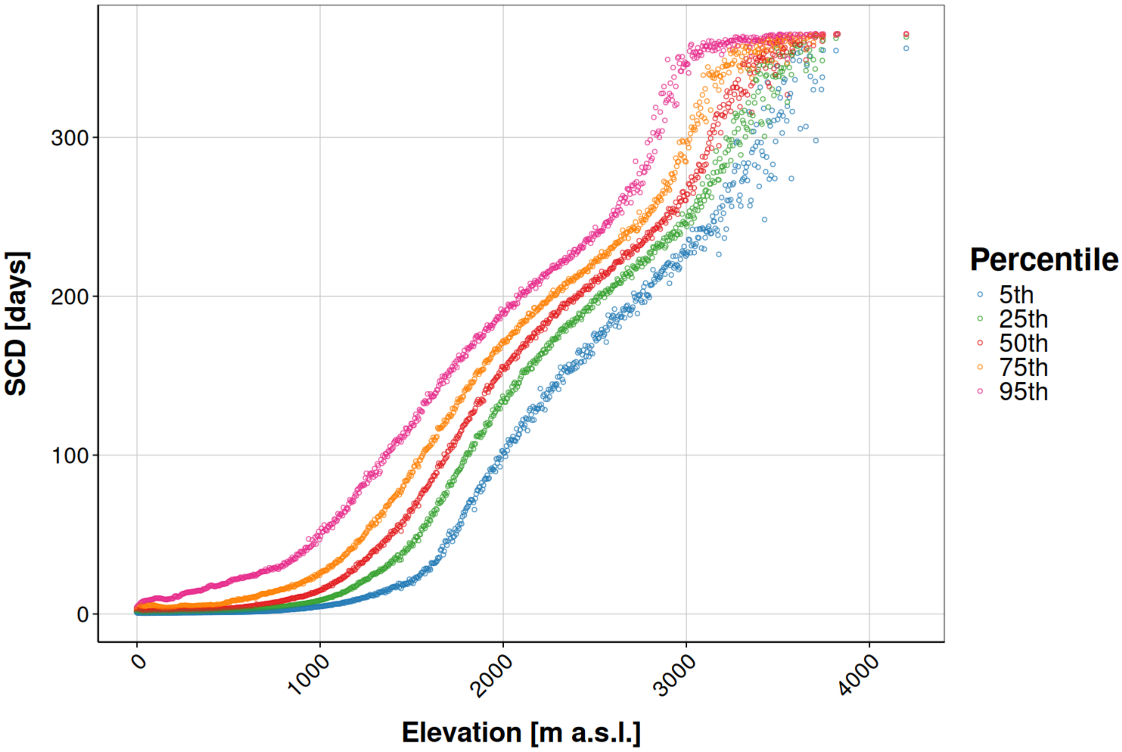

Figure 1.7: MODIS

1.4 Po Basin

This snow dataset focuses on the mountainous part of the Po River District, which is an over-regional entity resulting from the union of the Po River Basin with some smaller ones such as Conca and Reno basins. This territory, spanning eight regions of Northern Italy and parts of Switzerland and France, covers almost 87000 km^2. Assessing the amount of water stored as snow in the highlands of this region is particularly important, because the Po River Basin is home to roughly one-third of Italy's total population and produces almost 55% of Italy's hydropower. This snow product is a thirty years long SWE reconstruction, covering a period spanning from 1992 to 2021. For every hydrological year, SWE maps for the mountainous part of the Po River District are given on a daily basis during the period spanning from October 3^{rd} to July $1^{s}t$, which roughly corresponds to the period where snow coverage extends beside glacerized area. These SWE maps have a spatial resolution of 500 $\times$ 500 m^2 and are created using a hybrid modeling approach "J-snow".

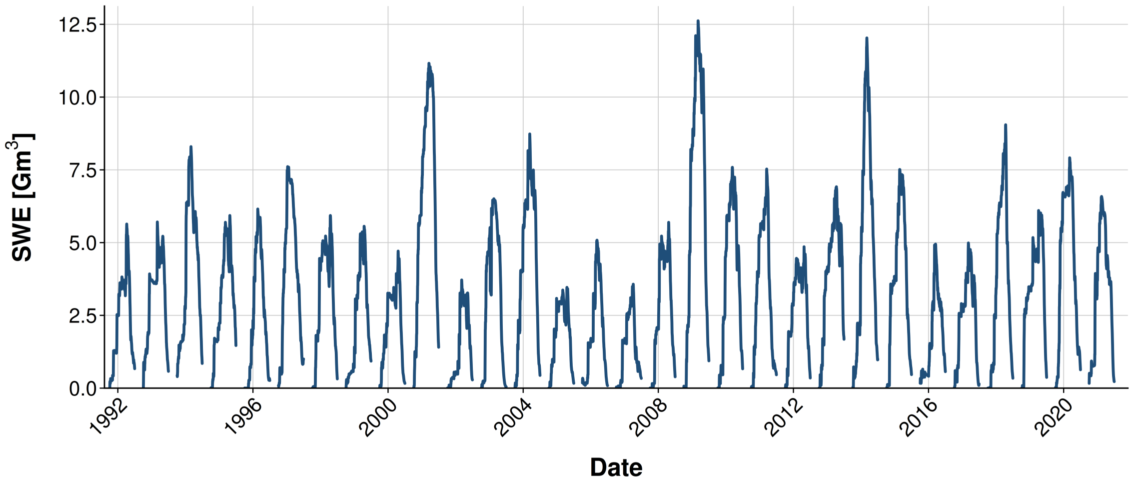

Figure 1.8: MODIS

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